

Hydrogen Fuel Assist for Marine Engines

Save fuel. Cut emissions. Extend engine life. Meet IMO targets. Using only water.

THE MARITIME OPPORTUNITY

The maritime industry faces mounting pressure from IMO regulations, rising fuel costs, and tightening environmental scrutiny. While much attention goes to large vessel propulsion and alternative fuel transitions, the immediate, high-impact opportunity lies in the thousands of auxiliary engines across all vessel classes and the propulsion engines of smaller working vessels — tugboats, DP ships, platform supply vessels, anchor handling tugs, fishing trawlers, coastal cargo vessels, ferries, and inland waterway craft.

These engines run continuously, burn enormous quantities of marine diesel (MDO), marine gas oil (MGO), and intermediate fuel oil (IFO). Even a modest 2–15% fuel saving per engine translates to significant annual cost reductions per vessel — and measurable CO2 reduction that directly improves CII ratings and IMO compliance.

THE SGT GREENMARINE SOLUTION

SGT HydroEdge has developed GreenMarine — a proprietary, patent-applied HHO (oxyhydrogen) generation system purpose-built for marine environments. GreenMarine generates hydrogen-oxygen gas on-demand through water electrolysis and injects it into the engine's air intake. The HHO acts as a powerful combustion catalyst — hydrogen's rapid flame propagation and extremely low ignition energy enable more complete combustion of any type of marine fuel.

The result: your engines burn less fuel for the same power output, emit dramatically fewer pollutants, and — critically — **experience continuous internal decarbonisation that extends engine life and reduces maintenance costs.**

TARGET ENGINES & VESSEL TYPES

Auxiliary Engines (All Vessel Classes)

On-board power generation DG sets on any vessel — from coastal cargo ships to large tankers and bulk carriers. These engines run continuously at port and at sea, making them ideal for HHO-assisted combustion.

Propulsion Engines (Smaller Vessels)

Medium-speed diesel propulsion on tugboats, DP vessels, PSVs, AHTSVs, trawlers, ferries, patrol boats, coastal and inland waterway vessels. These engines run on MDO/MGO and respond exceptionally well to HHO enhancement.

Tugboats

DP Ships & PSVs

Fishing Trawlers

Coastal Cargo

Ferries & RoPax

Inland Waterway

Patrol & Navy

AHTSVs

PROVEN RESULTS FROM DIESEL ENGINE RESEARCH BY SGT

3–15%

Fuel Savings

Up to 95%

PM & Smoke

>50%

CO Reduction

>46%

HC Reduction

3–15%

CO2 Reduction

Significant

NOx & SOx Reduction

THE HIDDEN BENEFIT: ENGINE DECARBONISATION & EXTENDED LIFE

SGT's research has revealed a critical additional benefit of HHO-assisted combustion in marine engines: continuous internal decarbonisation.

Marine diesel engines accumulate carbon deposits on pistons, cylinder heads, valves, injector tips, and turbochargers over thousands of operating hours. This carbon buildup reduces engine efficiency, increases fuel consumption, causes hot spots that damage components, and ultimately shortens engine life. Conventional solutions require engine downtime for manual carbon cleaning.

HHO gas, when introduced into the combustion chamber continuously, acts as a cleaning agent during normal operation. The oxygen-rich combustion environment burns off carbon deposits progressively, keeping engine internals cleaner without any downtime or manual intervention. This translates directly to:

- **Longer engine life** — reduced carbon buildup means less thermal stress, fewer hot spots, and slower component wear
- **Lower maintenance costs** — fewer carbon-related overhauls, cleaner injectors, less turbocharger fouling
- **Sustained performance** — engine efficiency does not degrade as rapidly over time, maintaining fuel savings throughout operational life
- **Reduced dry-docking time** — less engine maintenance required during scheduled dry-dock periods

SGT'S PATENT-APPLIED TECHNOLOGY ADVANTAGE

High HHO Yield	Patent-applied cell design delivers significantly higher gas output per unit of electrical input — critical for marine where power efficiency matters
Low Power Consumption	Optimised electrode geometry minimises electricity draw from vessel's power system
Marine-Grade Build	Corrosion-resistant materials and marinised enclosures designed for salt air, humidity, and vibration environments
Safety by Design	On-demand generation — zero hydrogen storage aboard. Flashback arrestors, pressure relief, and fail-safe controls. No SOLAS classification concerns for stored hydrogen.
Easy Maintenance at Sea	Modular design serviceable by ship's crew. No specialist technicians. Input: water and standard electrolyte.
Compact Footprint	Designed to fit in engine rooms of working vessels without requiring additional space or structural modification

DIGITALLY CONNECTED — IOT & DIGITAL TWIN READY

Every GreenMarine unit is fully digitally connected.

Built-in IoT sensors monitor system performance and engine parameters via satellite or cellular connectivity. Data feeds into GreenVision — SGT's AI-powered Digital Twin platform.

- **Voyage-level fuel tracking** — actual fuel savings quantified per voyage, per engine
- **Emission reporting** — CO₂, PM, CO, HC, SO_x, NO_x reductions documented for IMO DCS and CII compliance
- **Fleet dashboard** — monitor all equipped vessels from a single shore-based interface
- **Predictive maintenance** — detect engine and HHO system degradation before failures, reducing unplanned downtime

WHAT GLOBAL RESEARCH PROVES

While HHO application in marine engines remains a frontier area with limited direct maritime studies, the underlying diesel engine combustion science is extensively validated. Marine auxiliary engines and smaller vessel propulsion engines use the same fundamental combustion principles as industrial and automotive diesel engines — the same physics applies:

FUEL EFFICIENCY & ENGINE PERFORMANCE

- Fuel consumption reductions of 11.7–20% documented across multiple diesel engine studies, with 12.6–13.7% average BSFC improvement consistently reported
- Engine thermal efficiency improves by up to 31.5% with HHO supplementation — more energy extracted from each litre of fuel
- Brake power increases of 9–22.4% demonstrated, meaning engines can deliver the same output with less fuel, or more output with the same fuel
- 500-hour continuous dual-combustion study (diesel + HHO) confirmed no structural damage to engine components, validating long-term operational safety and reliability for continuous marine duty
- HHO has been validated as an auxiliary fuel and additive for marine turbines and diesel engines in published research, with application to the maritime sector specifically identified as a high-potential opportunity

EMISSION REDUCTIONS

- PM and smoke opacity reduced by 25–95% — the most dramatic single-intervention emission improvement available for diesel engines
- CO emissions reduced by 19.8–54.5%, HC emissions reduced by 7.7–46.15% — confirming significantly more complete fuel combustion
- CO₂ emissions reduced by 7–15.2% as a direct consequence of burning less fuel for equivalent power output — directly improving CII ratings
- SO_x reductions follow proportionally from fuel savings: less fuel burned = less sulphur released, complementing any fuel switching strategy
- NO_x reductions of 4.3–15.5% documented at moderate loads. SGT's optimised system calibration targets significant overall NO_x reduction

ENGINE DECARBONISATION & LONGEVITY

- HHO-assisted combustion burns off internal carbon deposits during normal engine operation — a process SGT has validated through its own research
- Cleaner combustion produces less soot and unburned carbon, reducing deposit formation rate on pistons, valves, injectors, and turbochargers
- Research confirms HHO addition shortens ignition delay and produces more complete combustion, leaving fewer carbon residues in the combustion chamber
- Reduced carbon buildup translates to lower thermal stress on engine components, fewer hot spots, and measurably slower wear rates

Research findings based on work by scientists at institutions including University of Technology Baghdad, UNAM Mexico, National University of Sciences and Technology Pakistan, multiple research groups in India, Turkey, Egypt, Malaysia, and published in ScienceDirect, Springer, MDPI, SAGE, Elsevier, and Nature Scientific Reports. SGT's own research on engine decarbonisation adds proprietary data. Full references available on request.

IMO REGULATORY CONTEXT

GreenMarine addresses multiple IMO and international maritime regulatory requirements simultaneously:

IMO GHG Strategy	40% carbon intensity reduction by 2030, net-zero ~2050. GreenMarine delivers immediate CO2 reduction.
CII Rating	Fuel savings directly improve CII score. Avoid D/E rating penalties.
EEXI Compliance	Documented efficiency improvements support EEXI compliance.
EU ETS (Shipping)	Less fuel = fewer carbon allowances required for EU voyages.
MARPOL Annex VI	PM, NOx, SOx reductions support ECA compliance.
IMO DCS	GreenVision provides verified fuel and emission data for reporting.

Target engines	Auxiliary engines (all vessels) + propulsion (smaller working vessels)
Fuel types	MDO, MGO, IFO, LSFO — any marine diesel fuel
Installation	Retrofit. No engine modification. Performed alongside scheduled maintenance.
Hydrogen storage	None — on-demand generation. No SOLAS storage concerns.
Input required	Water and electricity only. Serviceable by ship's crew.
Connectivity	IoT enabled. Satellite/cellular. GreenVision Digital Twin.
Technology	Patent-applied. Marine-grade. High yield. Low power.

Ready to Cut Your Fleet's Fuel Costs & Emissions?

Ask us for a no-obligation assessment of your vessel operations.

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SGT HydroEdge — Sustainability Infrastructure for Industry